

Title: Sea Buckthorn Berry: A Comprehensive Review of Its Therapeutic Potential and Stem Cell Mobilization Effects

Abstract:

Sea buckthorn (*Hippophae rhamnoides*) berry has garnered significant attention recently due to its rich nutritional profile and potential health benefits. This review aims to synthesize current research on sea buckthorn berry, focusing on its nutritional composition, therapeutic applications, and novel findings regarding its effects on stem cell mobilization.

Introduction

Sea buckthorn berry, derived from the sea buckthorn shrub, has been used for centuries in traditional medicine across Asia and Europe. Recent scientific investigations have begun to elucidate the mechanisms behind its purported health benefits.

Nutritional Profile

Sea buckthorn berries are exceptionally nutrient-dense, containing:

- **Vitamins:** High in vitamins C (up to 15 times more than oranges), A, E, K, and B complex
- **Minerals:** Rich in iron, calcium, magnesium, potassium, copper, and zinc
- **Fatty Acids:** Unique profile of omega-3, 6, 7, and 9 fatty acids, including rare omega-7 palmitoleic acid
- **Antioxidants:** High in flavonoids (quercetin, isorhamnetin), carotenoids (beta-carotene, lycopene, zeaxanthin), and phenolic compounds
- **Other Compounds:** Phytosterols, organic acids, amino acids, and dietary fiber

Therapeutic Applications

Cardiovascular Health

Sea buckthorn may improve cardiovascular function by:

- Reducing inflammation and improving blood flow
- Protecting against oxidative damage to blood vessels
- Potentially lowering blood pressure and cholesterol levels

Metabolic Health

Recent studies suggest potential benefits for diabetes and obesity management:

- Improving insulin sensitivity and glucose metabolism
- Supporting weight management efforts

Skin Health

The oil derived from sea buckthorn berries may benefit skin health by:

- Improving skin hydration and elasticity
- Potentially aiding in the treatment of various skin conditions, including atopic dermatitis and psoriasis
- Supporting wound healing and tissue regeneration

Anti-tumor Properties

Active components like isorhamnetin, quercetin, gallic acid, and protocatechuic acid demonstrate significant anti-tumor properties⁵.

Liver Protection

Sea buckthorn has shown hepatoprotective properties, potentially aiding in liver protection and regeneration.

Stem Cell Mobilization

A groundbreaking study by Drapeau et al. (2020) revealed that sea buckthorn berry extract can selectively mobilize certain types of stem cells²:

- Increased CD45dim CD34+ CD309- progenitor stem cells
- Boosted CD45- CD31+ CD309+ endothelial stem cells
- Elevated CD45- CD90+ lymphocytic mesenchymal stem cells

This selective mobilization could potentially enhance the body's regenerative and reparative functions.

Safety and Considerations

While generally considered safe when consumed as food, high doses may have pro-oxidant effects. Potential interactions with medications, particularly those affecting blood pressure or blood clotting, should be considered.

Conclusion

Sea buckthorn berry shows promising potential in various therapeutic applications, particularly in cardiovascular health, metabolic regulation, and stem cell mobilization. However, further research, especially in human clinical trials, is needed to fully elucidate its mechanisms of action and optimal therapeutic applications.

References

- 1 Teng et al. (2024). Research progress on antitumor effects of sea buckthorn. *Frontiers in Nutrition*.
- 2 Drapeau C et al. (2020). Rapid and selective mobilization of specific stem cell types after consumption of extract of sea buckthorn berries in healthy human subjects. *Clinical Interventions in Aging*; 14:253-263.
- 3 Gupta et al. (2005). Preclinical study of the effects of sea buckthorn leaf extracts on wound healing in albino rats.
- 4 Wang ZY, Luo XL, He CP. (2006). *Nan Fang Yi Ke Da Xue Xue Bao*. 26(1):124-5.
- 5 *Frontiers in Nutrition*. (2024). Research progress on antitumor effects of sea buckthorn.

Sea buckthorn (*Hippophae rhamnoides*) is a deciduous shrub renowned for its nutrient-rich berries, which have been utilized in traditional medicine and nutrition. Below is a comprehensive overview of its benefits, potential side effects, and considerations:

Benefits:

Benefit	Description
Rich Nutrient Profile	Sea buckthorn berries are abundant in vitamins C, E, K, and B12, as well as carotenoids like beta-carotene, zeaxanthin, and lycopene. They also contain essential minerals such as potassium, manganese, and copper. Wikipedia
Skin Health	The oil derived from sea buckthorn is rich in palmitoleic acid, which is beneficial for skin hydration and may aid in wound healing. It's commonly used in skincare products targeting dry or aging skin. Healthline
Anti-Inflammatory Properties	Sea buckthorn oil contains omega-7 fatty acids, known for their anti-inflammatory effects, which may support cardiovascular health and reduce chronic inflammation. Vogue
Cardiovascular Support	The phytosterols in sea buckthorn can help block cholesterol absorption, potentially reducing LDL (bad) cholesterol levels and supporting heart health. Healthline
Antioxidant Effects	High levels of antioxidants, including vitamins C and E, help combat oxidative stress, potentially reducing the risk of chronic diseases. PubMed Central
Digestive Health	Traditionally, sea buckthorn has been used to treat gastrointestinal disorders, though more research is needed to confirm these benefits. WebMD

Potential Side Effects and Considerations:

Consideration	Description
Blood Thinning	Sea buckthorn may slow blood clotting, increasing the risk of bleeding, especially when combined with anticoagulant medications. RxList
Blood Pressure	It might lower blood pressure, which could be problematic for individuals with hypotension or those on blood pressure-lowering medications. RxList
Blood Sugar Levels	Sea buckthorn may lower blood sugar levels, posing a risk for individuals with diabetes, especially those on glucose-lowering medications. WebMD
Allergic Reactions	Some individuals may experience allergic reactions, including rashes or gastrointestinal discomfort. WebMD
Pregnancy and Breastfeeding	There is insufficient reliable information about the safety of sea buckthorn during pregnancy and breastfeeding; it's advisable to avoid use during these periods. RxList
Surgical Considerations	Due to its potential blood-thinning effects, it's recommended to discontinue use at least two weeks before scheduled surgery to reduce bleeding risks. RxList

Conclusion:

Sea buckthorn berries offer a range of health benefits due to their rich nutrient and antioxidant content. However, potential side effects and interactions, especially concerning blood clotting, blood pressure, and blood sugar levels, should be carefully considered. It's advisable to consult with a healthcare professional before incorporating sea buckthorn into your regimen, particularly if you have underlying health conditions or are taking other medications. **Recent Insights on Omega-7 Fatty Acids and Sea Buckthorn**